

**DEPARTMENT OF CHEMICAL AND BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 331

BATCH DISTILLATION EXPERIMENT*

1. Introduction

Distillation is the oldest and most widely used separation process in industry. There are two primary distillation configurations, batch distillation and continuous distillation. Batch distillation is a transient process whereas continuous distillation is (or is close to being) a steady state process. Batch distillation is the simpler of the two processes because it requires much less control equipment. Continuous distillation is meant for high throughput separation, such as petroleum refining, whereas batch distillation is a low-throughput operation that is used mostly in the production of high-valued products, such as pharmaceuticals, biochemicals, and specialty chemicals where its higher unit separation cost can be justified. Because batch distillation requires much less control equipment, it is much more flexible than continuous distillation and a batch still can be easily cleaned and utilized for processing different mixtures.

In batch distillation, a liquid mixture of components to be partially separated is charged to a still and heated at a fixed pressure (typically 1 atm or lower). The more volatile component(s) appear preferentially in the vapor fraction while the less volatile component(s) remain preferentially in the liquid fraction. Condensation and removal of the vapor fraction permits a separation of the mixture components.

In this experiment, the separation characteristics of a methanol-isopropanol mixture will be studied in a batch distillation unit.

2. Objective

The objective of the experiment is to operate a batch distillation unit and to learn how to collect and analyze batch distillation data so that the characteristic features of the process can be quantified.

3. Theory

The Rayleigh equation, which is derived below, provides the basis for most of the data analysis on all batch distillation processes. As will be discussed in **DATA ANALYSIS**, the data analysis may be rigorous or approximate, depending on the assumptions which may be made, but the Rayleigh equation is often the starting point for the analysis.

Consider the transient material balance at time t on a batch distillation unit that contains a liquid solution that is boiling at a constant pressure P_0 . Let $T(t)$ and $N^L(t)$ denote the temperature and total mole number of the liquid solution remaining in the still at time t , $N_i^L(t)$ and $x_i(t)$ denote the corresponding mole number and mole fraction of component i in the liquid solution, and $dN^L(t)$, $dN^V(t)$ and $y_i(t)$ denote, respectively, the differential change in the total mole number of the liquid solution (in the still) and the corresponding differential mole number and mole fraction

*P. Lim, M. Cooper and K. Dickey, Department of Chemical and Biomolecular Engineering, NC State University, Raleigh, NC 27695, revised January 8, 2017.

of component i of the vapor stream leaving the still. All these variables vary with time as the batch distillation proceeds.

An overall material balance requires that

$$dN^L(t) + dN^V(t) = 0 \quad \text{or} \quad dN^V(t) = -dN^L(t) \quad (1)$$

A component i balance requires that

$$dN_i^L(t) = -y_i(t) * dN^V(t) \quad (2)$$

but also

$$dN_i^L(t) = d[N_L(t) * x_i(t)] = N_L(t) dx_i(t) + x_i(t) dN^L(t) \quad (3)$$

Combining Equations 2 and 3 give

$$dN_i^L(t) = N_L(t) dx_i(t) + x_i(t) dN^L(t) = -y_i(t) * dN^V(t) \quad (4)$$

Substituting Equation 1 into Equation 4 and rearranging the terms gives

$$N_L(t) dx_i(t) + x_i(t) dN^L(t) = y_i(t) * dN^L(t) \quad \text{or} \quad N_L(t) dx_i(t) = [y_i(t) - x_i(t)] dN^L(t) \quad (5)$$

By separating the variables in Equation 5 and integrating both sides of the resulting equation correspondingly, the Rayleigh equation is produced:

$$\int_{N_o^L}^{N^L(t)} dN^L(t) / N^L(t) = \ln[N^L(t) / N_o^L] = \int_{x_{io}}^{x_i(t)} dx_i(t) / [y_i(t) - x_i(t)] \quad (6)$$

where N_o^L is the total mole number of liquid solution initially charged to the still ($t = 0$) and x_{io} is the corresponding mole fraction of component i in the initial charge. Note that the $x_i(t)$ and $y_i(t)$ on the right side of Equation 6 are the **instantaneous** values of liquid and vapor mole fractions of component i at time t (see **DATA ANALYSIS**).

4. Experimental

4.1 Safety Concerns

1. Methanol and isopropanol are flammable and toxic. Wear gloves when handling the chemicals and avoid undue exposure. Clean up all spills and dispose of excess methanol and isopropanol in the waste container located at the experiment.

2. Avoid contact with the hot still while it is operating.

3. ***Never Operate the Still with the Heating Coil Fully Exposed, A Fire and Explosion Could Result!***

4.2 APPARATUS AND PROCEDURE

1. A schematic of the batch distillation unit and the control equipment is shown in Figure 1. Examine the distillation column and identify its key features, including the reboiler with its heating coil, the condenser and its cooling water inlet and outlet, the three trays below the condenser, the two sampling stems (only the one with the small condenser will be used in this experiment), the thermocouples at three different locations (only TC3 will be used to monitor the temperature of the solution in the pot during the experiment), the reflux switch valve, and the refill and drain valves on the still.

2. Examine the control panel and identify its key features, including the main power switch, the thermocouple switch, the heater power switch, the rotary Variac (which controls voltage to the heating coils and the heating rate), and the digital display windows for temperature, voltage and current readings.

3. There are five specific jobs which must be completed as distillation proceeds in order to collect all necessary data for the experiment, and each task should be the responsibility of a single team member. Before proceeding with the experiment, your team should agree on the job each member will perform. The jobs and their associated tasks are:

- Sample Collector – Collect samples of distillate as they emerge from the sampling stem.
- Timer – Record still temperature and heater voltage/current 1) as soon as the voltage switch is turned on, 2) every minute until distillate is formed, 3) after collection of each new distillate sample. Communicate with the sample collector during distillate collection

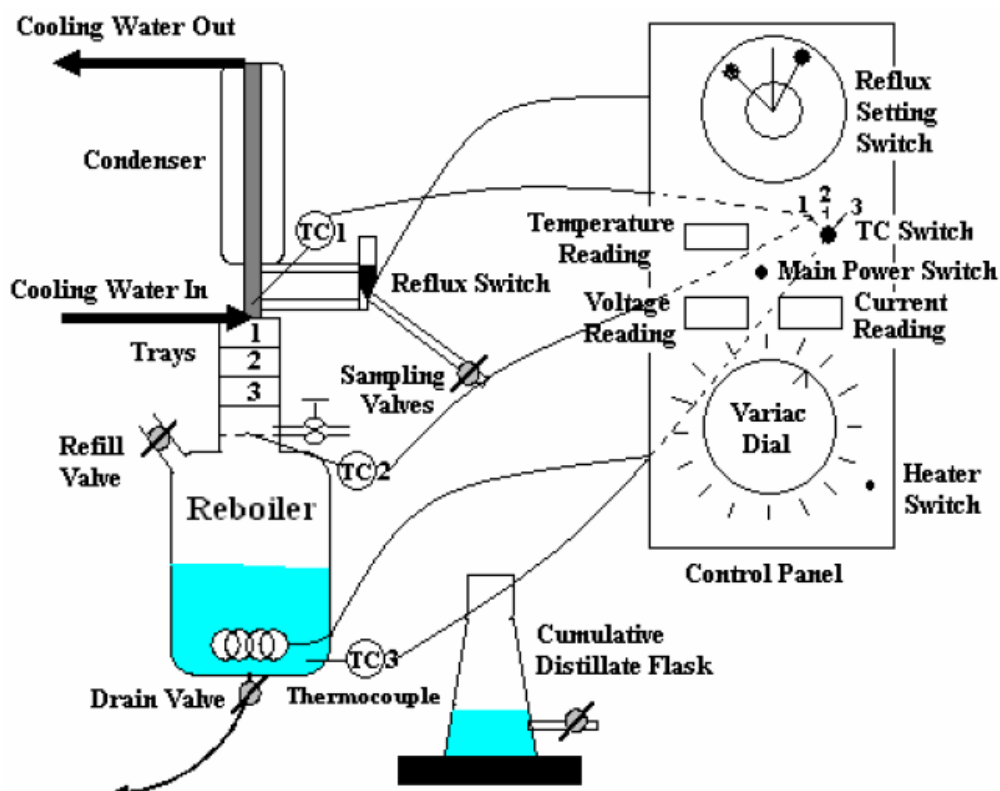


Figure 1. Simple schematics of the batch distillation column and controls.

- Mass Recorder – Record mass of samples as they are handed to you by the Sample Collector, before you give them to the Refractive Index Measurer.
- Refractive Index Measurer – Perform initial calibration of refractometer at the beginning of the laboratory, then determine refractive index of each sample using the refractometer. After measuring the refractive index for each sample, pour the sample into the cumulative distillate flask (a large beaker), this should be done near the end of each run to minimize evaporative losses. Record refractive index of each sample.
- Spreadsheet Recorder - Put mass and refractive index data into the spreadsheet. It is also your job to begin producing the spreadsheet for data analysis.

4. Before proceeding with the experiment, become familiar with use of the refractometer and perform its initial air/water calibration by following the instructions in Section 4.3. Prepare a set of labeled glass bottles for collecting instantaneous distillate samples. Make sure that the bottles are clean and dry. The masses of each of the empty sample bottles should also be recorded at this time.

5. After calibrating the refractometer with air and water, calibration curve must then be produced which relates refractive index to wt% methanol in the methanol/isopropanol solution. Following the procedure described in Section 4.3.2, record the refractive indexes for standard solutions of 0 wt%, 25 wt%, 50 wt%, 75 wt% and 100 wt% methanol to produce the calibration curve. Plot your data and confirm that the refractive index of a methanol-isopropanol mixture is linearly dependent on the weight-fraction composition. Determine the least-square fit for your calibration data and use the resulting equation to determine the composition of the distillate samples from your experiment. This task should be completed while the still is heating up.

6. The initial charge to the still should be approximately 4 L of a roughly 40 wt% methanol in isopropanol solution. If the TA has not already placed the charge into the still at the beginning of the experiment, mix approximately 1600 mL (1269.2 g) of methanol and 2400 mL (1908.4 g) of isopropyl alcohol and pour into the reboiler through the top refill valve on the still.

7. Slowly drain the initial solution in the reboiler through the drain valve into a beaker and measure the mass of the solution (don't forget to record the mass of the empty beaker). Withdraw representative samples of the solution and use the refractometer to determine its mass-fraction composition. These measurements are **essential** to your data analysis: mass and composition of the initial charge will be needed for the required material balances. Close the drain valve and return the solution to the reboiler through the top refill valve on the still.

8. Turn on the water supply line to the condenser by turning the yellow handled valve behind the column. (The TA should have pre-adjusted the water flow for sufficient cooling capability) Why is it important to turn on the cooling water first before turning on the heating coil in the still?

9. Set the desired reflux ratio for the run by adjusting the needles on the dial of the reflux unit, then turn on the reflux power switch. The reflux ratio on this still is based on a timing mechanism which activates a valve that directs condensate either back to the reboiler or through

the sampling stem. The needles on the dial correspond to the reflux ratio (RR) as described in Equation 7:

$$RR = \frac{Needle_{right}}{Needle_{left}} \quad (7)$$

For instance, setting the needle on the right side of the dial to 15 and the needle on the left side of the dial to 10 would give a reflux ratio of 1.5. Two batch runs should be performed at two sufficiently different reflux ratios (e.g., $RR_1 = 0.75$ and $RR_2 = 2R_1 = 1.5$) to ensure a clear-cut determination of the effect of reflux ratio on the separation.

10. Plug in the 3-phase, 220-V plug into the wall socket to connect electrical power to the batch distillation equipment. Turn on the main power switch located in the middle of the control panel.

11. Set the Variac to the desired voltage such that the voltmeter reading would be less than 220V. Turn on the power and *simultaneously start a stopwatch. This is the start of the run.* (The start does NOT correspond to the appearance of the first condensate!). Monitor and record the voltage, current and pot temperature readings as a function of time during the run.

12. Prepare the cumulative distillate flask for holding each instantaneous condensate sample after its refractive index has been recorded. Place a magnetic stir bar in the flask and place it on the magnetic stirring pad.

13. Note the time when distillate first drips from the sampling stem and collect samples of instantaneous distillate with the dry bottles (this should take 10-20 min). For each sample collected, record the collection time and its mass (which can be determined by weight difference), and determine its refractive index. Sampling of the instantaneous distillate should be more frequent at the start of each distillation run—about once every minute in the first eight minutes—but thereafter the sampling period can increase progressively with time. Suggested sampling times are: 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 20, 23, 26, 29, 32, 35 minutes after the first drop of distillate appears. You may find it helpful to start a second timer at this point to track sampling time separately from overall trial run time. The numbered labels on the glass bottles correspond to the elapsed time in the experiment when samples should be taken. **Note:** *If any student finds that they are sensitive to the fumes from the distillation column, let the TA or Dr. Golpour know.*

14. After each instantaneous condensate sample has been analyzed for its refractive index, pour what remains of the sample into the cumulative collection vessel. Exercise care in the transfer operation to avoid spills or splashes when the magnetic stirrer turned on.

15. The run can be stopped after sufficient data have been collected over a run time of about 35 minutes or if the heating coil becomes exposed, in which case the run should be stopped **immediately**. After the first run has been completed, turn off the heater power, and reflux control (this lowers the reflux switch) and allow the still to cool down. While the still is cooling, measure the final mass and refractive index of the cumulative distillate to provide a material balance check as part of your data analysis.

16. After measuring the mass and refractive index of the drained solution from the still, mix the drained solution with the cumulative distillate and measure the total mass and refractive index of the resulting solution. These measurements provide the initial starting conditions for the second distillation trial.

17. Carefully transfer the resulting solution to the still and perform a second run with the second reflux ratio. Return to Step 9 and repeat Steps 9-15 to complete a batch distillation for the second reflux ratio. Upon completion of the second run, turn off the main power switch, the heater switch, the reflux control, and the cooling water valve. Return all solution back to the reboiler through the top refill valve on the still. ***Note: Groups should consider rotating tasks for the second reflux ratio to gain additional hands-on experience.***

4.3 Refractometer Use

Since refractive index varies as a function of temperature, the refractometer used in this experiment utilizes Peltier thermoelectric modules to ensure the refractive indexes for all samples are recorded at the same temperature (25°C). A calibration with air and water must be performed when first turning on the refractometer at the beginning of the experiment. Then, a calibration curve will be produced which demonstrates the relationship between refractive index and mass fraction methanol in the binary methanol/isopropanol system by determining refractive indexes for 0, 25, 50, 75 and 100 wt% methanol in isopropanol samples.

4.3.1 Initial Calibration

1. Turn on the refractometer with the power button on the front of the device. Once the home screen appears, lift the lid of the device and add a few drops of pure isopropanol (given in a labeled vial on the lab bench) to the bowl, enough to cover the gray sensor. Wipe out the bowl and sensor with a Kimwipe; do NOT use a tissue, towel, etc., which may scratch the delicate sensor.
2. On the home screen, press the “25°C” button, then press “Start” on the following screen. Wait at the next screen for the sensor temperature to stabilize, which may take a few minutes.
3. After reaching and maintaining temperature, the refractometer will automatically begin recording the refractive index for air. Wait for the air calibration to complete.
4. Once the air calibration is complete, the system will prompt you to “Add sample.” Open the lid of the refractometer and place a few drops of water from the provided squeeze bottle onto the sensor. Shut the lid, press “OK” and wait for the device to complete the measurement.
5. When prompted to “Drain cell,” open the lid and wipe the sensor with a Kimwipe. When prompted to “Rinse cell with water,” press “OK.”
6. At the “Rinse cell with acetone” prompt, add a few drops of pure isopropanol on top of the sensor, then press “OK.”
7. At the “Dry cell” prompt, wipe out the bowl with a Kimwipe to remove the isopropanol. Shut the lid and press “OK.”

8. At this point the refractometer will indicate if the initial calibration passed or failed. If the calibration passed, continue by determining refractive indexes for the 0, 25, 50, 75 and 100 wt% methanol in isopropanol samples. If calibration failed, repeat Steps 2-7 before proceeding.

4.3.2. Refractive index of samples

1. To measure the refractive index for a sample, begin by pressing the “RI°25” button on the home screen.
2. At the “Add sample” prompt, lift the lid of the apparatus and add a few drops of the sample to the bowl, enough to cover the sensor. Shut the lid and press “OK.”
3. At the “Drain cell” prompt, lift the lid and use a Kimwipe to wipe the sample from the bowl and sensor. Shut the lid and press “OK.”
4. Record the RI displayed on the screen, then press “OK.”
5. Lift the lid and add a few drops of isopropanol to the sensor, then wipe out the bowl and sensor with a Kimwipe.
6. Repeat Steps 1-5 for each sample.

NOTE: The methanol weight percent calibration curve is generated by measuring the standard samples in this manner and then generating a curve in Excel, NOT by using the calibration mode in the refractometer.

DATA ANALYSIS

The batch distillation data should be analyzed in different ways as follows: (A) rigorous Rayleigh analysis using only collected data and a material balance, (B) approximate Rayleigh analysis based on a one-stage equilibrium assumption based only on VLE data, (C) approximate Rayleigh analysis based on a one-stage equilibrium assumption in conjunction with a constant volatility assumption, (D) McCabe-Thiele point-by-point analysis that is based on the assumptions of a multi-stage equilibrium and a pseudo steady state at each data point, and (E) Matlab simulation that is based on the assumptions of a one-stage equilibrium and the van Laar model for the activity coefficients of the constituent components in the solution. Be sure to discuss the effects of reflux ratio on the distillation as part of your data analysis. **Note 1:** the refractometer used in the laboratory is used to determine methanol wt% in each sample; the various Rayleigh analyses are calculated on mol% basis. **Note 2:** DO NOT attempt to perform error analysis for this experiment.

(A) Rigorous Rayleigh Analysis

The left-hand side of Equation 6 may be determined from the data collected on the solution in the still at the beginning and end of the run:

$$\ln \frac{N_f^L}{N_o^L} = \ln \frac{M_f \left[\frac{w_{Mf}}{MW_M} + \frac{1-w_{Mf}}{MW_I} \right]}{M_o \left[\frac{w_{Mo}}{MW_M} + \frac{1-w_{Mo}}{MW_I} \right]} \quad (8)$$

where N_o^L and N_f^L are the mole numbers of the solution in the still at the start and end of the run, respectively, M_o and M_f are the weights of the solution in the still at the start and end of the run,

respectively, MW_M and MW_I are the molecular weights of methanol and isopropanol, respectively, and w_{M_o} and w_{M_f} are the methanol weight fractions of the solution in the still at the start and end of the run, respectively.

The right-hand side of Equation 6, $\int_{x_{M_o}}^{x_{M_f}} \frac{1}{y_M - x_M} dx_M$, may be evaluated graphically or

numerically by plotting $\frac{1}{y_M(t) - x_M(t)}$ versus $x_M(t)$ as methanol *mole* fraction $x_M(t)$ varies from

x_{M_o} at the start of the run to x_{M_f} at the end of the run. Note that for a total condenser, the methanol mole fraction in the vapor phase at time t , $y_M(t)$, may be set equal to the methanol mole fraction of the corresponding instantaneous condensate sample taken at time t , $x_M^D(t)$, i.e., $y_M(t) = x_M^D(t)$. The methanol *mole* fraction of the solution remaining in the still at time t , $x_M(t)$, is not directly measured during the run, except at the beginning and end of the run. However, it may be determined by performing a running component balance on methanol and isopropanol after the withdrawal of each instantaneous condensate sample:

$$x_{M_n}(t) = \frac{\frac{1}{MW_M} [M_o w_{M_o} - \sum_{i=0}^{n(t)} m_i w_{M_i}]}{\frac{1}{MW_M} [M_o w_{M_o} - \sum_{i=0}^{n(t)} m_i w_{M_i}] + \frac{1}{MW_I} [M_o (1 - w_{M_o}) - \sum_{i=0}^{n(t)} m_i (1 - w_{M_i})]} \quad (9)$$

In Equation 9, the summation terms include all the n instantaneous distillate samples withdrawn from the still up to the time t , $x_{M_n}(t)$ is the corresponding methanol mole fraction of the solution remaining in the still after the n^{th} instantaneous condensate sample is drawn at time t , m_i is the weight of the i^{th} instantaneous distillate sample withdrawn, w_{M_i} is the corresponding methanol weight fraction of the i^{th} instantaneous distillate sample, and the other variables are as defined previously.

Thus, knowing $\frac{1}{y_M(t) - x_M(t)} = \frac{1}{x_M^D(t) - x_M(t)}$ and $x_{M_n}(t)$ as a function of the run time t , one

should be able to evaluate the right-hand side of Equation 6, either numerically or graphically (using the trapezoid or Simpson rule or by counting squares). The result may then be compared with the left hand side of Equation 6. An agreement within $\pm 10\%$ would indicate that the data are consistent.

The data taken on the cumulative distillate provides an independent check on the material balance. Instead of Equation 9, the methanol mole fraction of the solution remaining in the still may alternatively be determined from the data taken on the cumulative distillate, i.e.,

$$x_{M_n}(t) = \frac{\frac{1}{MW_M} [M_o w_{M_o} - M^{\text{TotD}} w_M^{\text{TotD}}]}{\frac{1}{MW_M} [M_o w_{M_o} - M^{\text{TotD}} w_M^{\text{TotD}}] + \frac{1}{MW_I} [M_o (1 - w_{M_o}) - M^{\text{TotD}} (1 - w_M^{\text{TotD}})]} \quad (9')$$

In Equation 9', M^{TotD} is the weight of the cumulative distillate collected up till time t (including the n^{th} instantaneous distillate sample), w_M^{TotD} is the corresponding methanol weight fraction of

the cumulative distillate up to that time, and the other parameters are as defined previously. Equations 9 and 9' should give the same results if there is no loss of material in sampling and analysis. Since some loss of material is inevitable in experimentation, Equation 9' may be expected to be less precise than Equation 9. Nevertheless, Equation 9' can provide a useful material balance check on the results given by Equation 9 if there is minimal loss of material in analysis.

(B) Approximate Rayleigh Analysis Based on a One-Stage Equilibrium Assumption via only VLE data

If the distillate may be assumed to be in equilibrium at all times with the solution remaining in the still (the one-stage equilibrium assumption), then the right-hand side of Equation 6 may be evaluated in two alternate ways. The first alternate way is to determine $x_M(t)$ from VLE data, such as that provided in Table 1. Knowing $y_M(t)$ as the measured $x_M^D(t)$ of the instantaneous distillate sample at time t , the equilibrium y_M versus x_M plot of the VLE data can be used to determine the corresponding value for $x_M(t)$, instead of using Equation 9 that is based on a rigorous mole balance on methanol and isopropanol. Then a series of $x_M(t)$ values corresponding to the series of measured $x_M^D(t)$ values can be generated for the instantaneous distillate samples taken over the course of the run, and used to evaluate the right-hand side of Equation 6 in a similar way as in the Rigorous Rayleigh analysis. The relative agreement between the two sides will determine the relative consistency of the data and the validity of the one-stage equilibrium assumption.

Table 1. VLE data (expressed in terms of methanol mole fractions) on the methanol (M)-isopropanol (I) system at 1.0 atm [from *Ind. Eng. Chem.*, 44, p. 2450 (1932)].

x_M	0.000	0.0810	0.1950	0.2930	0.4080	0.5220	0.6606	0.7900	0.9019	1.000
y_M	0.000	0.1320	0.2960	0.4285	0.5700	0.6850	0.8000	0.8910	0.9535	1.000
$T, ^\circ\text{C}$	82.4	81.0	78.9	77.1	74.8	72.7	70.2	67.9	66.2	64.96

(C) Approximate Rayleigh Analysis Based on a One-Stage Equilibrium Assumption and constant relative volatility

The second alternate way of evaluating the right-hand side of Equation 6 in conjunction with the one-stage equilibrium assumption is to assume a constant relative volatility assumption. The relative volatility of the methanol-isopropanol pair is defined by

$$\alpha_{MI} = \frac{K_M}{K_I} = \frac{y_M/x_M}{y_I/x_I} \quad (10)$$

For a binary mixture,

$$y_I = 1 - y_M \text{ and } x_I = 1 - x_M \quad (11)$$

hence

$$y_M = \frac{\alpha_{MI} x_M}{1 - x_M (1 - \alpha_{MI})} \quad (12)$$

and

$$\frac{1}{y_M - x_M} = \frac{1}{(\alpha_{MI} - 1)} \left(\frac{1}{x_M} + \frac{1}{1 - x_M} \right) + \frac{1}{1 - x_M} \quad (13)$$

It follows that for the assumption of a constant relative volatility (α_{MI}) the right-hand side of Equation 6 may be integrated analytically and Equation 6 itself may be shown to take on the closed form

$$\ln \frac{N_f^L}{N_o^L} = \frac{1}{(\alpha_{MI} - 1)} \ln \frac{x_{Mf}(1 - x_{Mo})}{x_{Mo}(1 - x_{Mf})} + \ln \frac{1 - x_{Mo}}{1 - x_{Mf}} \quad (14)$$

For analysis of the batch distillation data with Equation 14, the constant α_{MI} in the equation may be taken to be the mean α_{MI} value given by the VLE data over the composition range of interest. The relative agreement between the left- and right-hand sides of Equation 14 will determine the relative consistency of the data and the validity of the one-stage and constant volatility assumptions.

(D) McCabe-Thiele Point-by-Point Analysis Assuming a Multi-Stage Equilibrium and a Pseudo Steady State

If a multi-stage equilibrium is assumed along with a pseudo steady-state assumption, then the McCabe-Thiele analysis, which assumes an equimolal overflow, may be extended to the batch distillation data point by point. Recall that, on a y_1 vs x_1 plot with a 45°-line, the operating line for the enriching section of a continuous distillation column (above the feed stage) is given by

$$y_{M(n+1)} = \frac{RR}{RR+1} x_{M(n)} + \frac{1}{RR+1} x_M^D = \frac{RR}{RR+1} x_{M(n)} + \frac{1}{RR+1} y_M^D \quad (15)$$

where $y_{M(n+1)}$ = methanol mole fraction of the vapor stream leaving the (n+1)th stage (counting from the top) and rising to the nth stage directly above it, $x_{M(n)}$ = methanol mole fraction of the liquid stream leaving the nth stage and falling to the (n+1)th stage directly below it, RR = reflux ratio, and $x_M^D = y_M^D$ = methanol mole fraction of the instantaneous distillate. Starting with the point on the 45°-line that corresponds to methanol mole fraction of the instantaneous distillate (x_M^D or y_M^D) the operating line is drawn passing through the point and having a slope equal to $RR/(RR + 1)$. The total number of equilibrium stages—4 in our experimental setup—is stepped off, starting with the point on the 45°-line and then alternating between the equilibrium and operating lines. The methanol mole fraction of the solution in the still should then be given by x_{M4} which is associated with the 4th stage (again, counting from the top). The value of x_{M4} should agree with the corresponding x_M calculated using Equation 9, if the data are consistent.

Perform the McCabe Thiele analysis on two data points taken on each run. Pick any two instantaneous samples of distillate taken about one-third way, half way, or two-third way into the run. Starting with each measured composition of the instantaneous distillate sample and the known reflux ratio of the batch distillation run, construct a McCabe-Thiele plot on the y_M versus x_M diagram drawn from the data given in Table 1. Step off the four equilibrium stages and determine graphically the composition of the liquid mixture remaining in the still at the instant the instantaneous distillate sample was collected. Compare this graphically determined composition value with the corresponding composition value determined by Equation 9. How close is the agreement between the two values? List possible causes for any discrepancy.

(E) Matlab Simulation Assuming One-Stage Equilibrium & Van Laar Activity Coefficients

On the basis of material and energy balances, one can derive the following system of differential equations which governs the temperature T , total mole number N_L , and methanol mole fraction x_M of a methanol-isopropanol solution that is undergoing a batch distillation.

Before boiling occurs or for temperature $T(t) < T_B^0$, bubble point of the solution initially charged to the still,

$$\left(\frac{\partial N_L}{\partial t}\right)_P = 0 \quad \text{where } N_L \text{ is the total mole number of the solution in the still at time } t \quad (\text{D-1})$$

$$\left(\frac{\partial x_M}{\partial t}\right)_P = 0 \quad \text{where } x_M \text{ is the methanol mole fraction of the solution in the still at time } t \quad (\text{D-2})$$

$$\left(\frac{\partial T}{\partial t}\right)_P = \frac{Q_o}{N_L C_p} \quad (\text{D-3})$$

where $C_p = x_M^* C_{pM} + (1 - x_M) C_{pI}$, $C_{pM} = C_p$ of methanol, $C_{pI} = C_p$ of isopropanol, Q_o = rate of heat supplied to the batch still, assumed to be a constant for simplicity.

After boiling starts or for $T(t) \geq T_B$:

$$\left(\frac{\partial N_L}{\partial t}\right)_P = \frac{Q_o}{x_M C_p \left(\frac{\partial T}{\partial x_M}\right) \left(\frac{\gamma_M P_M^{sat}}{P^{Tot}} - 1\right) - \Delta H_{vap}} \quad (\text{D-4})$$

where $\Delta H_{vap} = x_M \Delta H_{vapM} + (1 - x_M) \Delta H_{vapI}$, ΔH_{vapM} = heat of vaporization of methanol, and ΔH_{vapI} = heat of vaporization of isopropanol. Also:

$$\left(\frac{\partial x_M}{\partial t}\right)_P = \frac{x_M}{N_L} (\gamma_M P_M^{sat} - 1) \left(\frac{\partial N_L}{\partial t}\right)_P \quad (\text{D-5})$$

$$\left(\frac{\partial T}{\partial t}\right)_P = \left(\frac{\partial T}{\partial x_M}\right)_P \left(\frac{\partial x_M}{\partial t}\right)_P \quad (\text{D-6})$$

$$\text{where } \left(\frac{\partial T}{\partial x_M}\right)_P = \frac{P-Q}{R}, \text{ and} \quad (\text{D-7})$$

$$P \equiv \gamma_M P_M^{sat} \left(\frac{2\alpha^2 x_M \left[1 + \frac{x_M}{1-x_M} \right]}{\beta(1-x_M) \left[1 + \frac{\left(\frac{\alpha}{\beta}\right)x_M}{1-x_M} \right]^3} - 1 \right) \quad (D-8)$$

$$Q \equiv \gamma_I P_I^{sat} \left(\frac{2\beta^2(1-x_M) \left[1 + \frac{1-x_M}{x_M} \right]}{\alpha x_M \left[1 + \frac{\beta(1-x_M)}{\alpha x_M} \right]^3} - 1 \right) \quad (D-9)$$

$$R \equiv \gamma_M x_M P_M^{sat} \frac{B_M}{(T+C_M)^2} + \gamma_I (1-x_M) P_I^{sat} \frac{B_I}{(T+C_I)^2} \quad (D-10)$$

where A_M , B_M , C_M are the Antoine parameters for methanol in $\log_{10}[P^{sat}/\text{mm Hg}] = A_M - B_M/(T/^\circ\text{C} + C_M)$, A_I , B_I , C_I are the Antoine parameters for isopropanol in $\log_{10}[P^{sat}/\text{mm Hg}] = A_I - B_I/(T/^\circ\text{C} + C_I)$, α and β are the parameters in the van Laar equations for activity coefficients of methanol and isopropanol, γ_M and γ_I :

$$\ln(\gamma_M) = \frac{\alpha}{\left[1 + \frac{\alpha x_M}{\beta(1-x_M)} \right]^2} \quad (D-11)$$

$$\ln(\gamma_I) = \frac{\beta}{\left[1 + \frac{\beta(1-x_M)}{\alpha x_M} \right]^2} \quad (D-12)$$

where $\alpha = -0.1354$ and $\beta = -0.107$, with the initial conditions: At $t = 0$, $N_L = N_{L0}$, $x_M = x_{M0}$ and $T = T_0$.

Physical Parameters: $A_M = 8.0868$, $B_M = 1584.02$, $C_M = 239.38$; $C_{pM} = 23.0 \text{ cal}/(\text{g-mol.K})$, $\Delta H_{\text{vapM}} = [9207 - 9(T/^\circ\text{C} - 25)] \text{ cal/g-mol}$; $A_I = 8.1430$, $B_I = 1583.43$, $C_I = 218.62$; $C_{pI} = 37.1 \text{ cal}/(\text{g-mol.K})$, $\Delta H_{\text{vapI}} = [10064 - 15.8(T/^\circ\text{C} - 25)] \text{ cal/g-mol}$

Equations D-1 through D-12 may be solved numerically using Matlab. Two Matlab M-files (BatchDistillation.m and BatchDistillationdv.m) are provided on the CHE 331 Moodle website which can be run on Matlab to generate the T , N_L and x_M profiles as a function of time for the following set of data: $N_{L0} = 100 \text{ g-mol}$, $x_{M0} = 0.70$, $T_0 = 25^\circ\text{C}$, $P_{\text{atm}} = P_{\text{Tot}} = 1 \text{ atm}$, $Q_0 = 12 \text{ kcal/min}$, T_B° (bubble point of the initial charge with $x_{M0} = 0.70$) = 69.63°C (Can you verify by a bubble point calculation?), Distillation Time Interval = 0-50 min.

Test run the Matlab program by copying and running M-files on Matlab, which you can access on any NT workstation on NCSU campus and through the VCL. You should be able to find Matlab on the Application Launcher. Double click the Matlab icon to invoke the Matlab command window and click the File pulldown menu to copy the two M-files into Matlab's working directory. Then on the command line (characterized by $>>$) enter BatchDistillation (without the .m extension). If the Matlab program runs successfully, you should see 3 plots on a single figure appearing on the screen. If there is any problem, Matlab will provide error message(s) on screen to help debugging the problem. You can use the M-files as a template and


```

%
%                               A*xM
%
% R = gM*xM*PsatM*BM/(T + CM)^2 + gI*(1 - xM)*PsatI*BI/(T + CI)^2
%
% with the initial conditions: At time 0, NL = NLo,  xM = xMo  and  T = To
%
% Parameter values: NLo = 100 g-mol,  xMo = 0.70,  PTot = 1 atm,  To = 25oC
%                      TBuo = 69.63oC,  Qo = 12.0 kcal/min
%
% log10[P/mm Hg] = A - B/(T/oC + C)  Antoine Coeff      A      B      C
%                               Methanol      8.0868      1584.02      239.38
%                               Isopropanol     8.1430      1583.43      218.62
%
%                               Heat capacity Cpi      Heat of vaporization Hvapi
% Methanol      23.0 cal/(g-mol.K)      9207 - 9(T/oC - 25) cal/g-mol
% Isopropanol   37.1 cal/(g-mol.K)      10064 - 15.8(T/oC -25) cal/g-mol
%
% van Laar model for activity coefficients
%
%                               A                               B                               A = -0.1354
% ln(gM) = ----- ln(gM) = -----
%                               A*xM 2                               B*xI 2                               B = -0.1075
%                               [1 + ----]                               [1 + ----]
%                               B*xI                               A*xM
%
% clc
% clf
% clear
% global CpM CpI HvapM HvapI Qo To TBuo PTot AM BM CM AI BI CI A B
%
% A = -0.1354;
% B = -0.1075;
%
% CpM = 23.0;
% CpI = 37.1;
%
% AM = 8.0868;
% BM = 1584.02;
% CM = 239.38;
%
% AI = 8.1430;
% BI = 1583.43;
% CI = 218.62;
%
% NLo = 100;
% xMo = 0.70;
% To = 25.0;
%
% PTot = 1.0;
% TBuo = 69.63;
% Qo = 12000;
%
% to = 0;
% tf = 50;
% tspan = [to tf];
% Y0 = [NLo xMo To];
%
% Solve the above differential equations numerically using the "ode15s"
% solver. The results are stored in a three-column matrix Y with the liquid

```

```

% mole number stored in the first column, the methanol mole fraction stored
% in the second column, and the solution temperature stored in the third
% column. The derivative functions of the differential equations are
% defined in a user-created file called BatchDistillationdv.m. Here t is a
column
% vector containing the incremental time in min and Y is a three-column
% matrix containing NL (mole number of the methanol solution remaining in
% the still) in the first column, xM (the methanol mole fraction of the
% solution in the still) in the second column, and T (the temperature of
% the methanol solution in the still) in the third column.

[t, Y] = ode15s('BatchDistillationdv', tspan, Y0);

% Plot the Matlab results on a single figure.

plot (t, Y(:,1), '+', t, 100*Y(:,2), '* ', t, Y(:,3), 'o'), title ...
    'Batch Distillation--NL, 100*xM, and T vs. Time',...
    legend ('Liquid Moles = Y(:,1)', 'MeOH Mole % = 100*Y(:,2)',...
        'Temp = Y(:,3)'),...
    xlabel ('t(min)'), ylabel ('NL(g-mol), 100*xM, T(oC)');
% Matlab derivative function file BatchDistillationdv.m which defines the
following
% system of differential equations that is to be solved numerically for
% the simulation of the batch distillation of aqueous methanol solution:
%
% For T(t) < TBuo (bubble point of the solution charged to the still),
%
%
%      dNL
%      --- = 0
%      dt
%
%      dxM
%      --- = 0
%      dt
%
%      dT      Qo
%      -- = ----- where Cp = xM*CpM + (1 - xM)CpI
%      dt      NL*Cp
%
% For T(t) = or > TBuo
%
%
%      dNL      Qo
%      --- = -----
%      dt      xM*Cp(---) (----- - 1) - Hvap
%                  dxM      PTot
%
%      where Hvap = xM*HvapM + (1 - xM)HvapI
%
%      dxM      xM      dNL
%      --- = -- (gM*PsatM - 1) ---
%      dt      NL      dt
%
%      dT      dT      dxM      dT      P - Q
%      -- = (---) (---) where --- = -----
%      dt      dxM      dt      dxM      R
%

```

```

%
% 
$$P = g_M \cdot P_{satM} \left\{ \frac{2A^2 x_M / B}{(1 - x_M)^2 \left[ 1 + \frac{A \cdot x_M}{B \cdot (1 - x_M)} \right]^3} - 1 \right\}$$

%
%
%
%
% 
$$Q = g_I \cdot P_{satI} \left\{ \frac{2B^2 (1 - x_M) / A}{x_M^2 \left[ 1 + \frac{B \cdot (1 - x_M)}{A \cdot x_M} \right]^3} - 1 \right\}$$

%
%
%
% 
$$R = g_M \cdot x_M \cdot P_{satM} \cdot B_M / (T + C_M)^2 + g_I \cdot (1 - x_M) \cdot P_{satI} \cdot B_I / (T + C_I)^2$$

%
function dY = BatchDistillationdv(t,Y)
global CpM CpI HvapM HvapI Qo To TBuo PTot AM BM CM AI BI CI A B

Cp = Y(2)*CpM + (1 - Y(2))*CpI;
HvapM = 9207 - 9*(Y(3) - 25);
HvapI = 10064 - 15.8*(Y(3) - 25);
Hvap = Y(2)*HvapM + (1 - Y(2))*HvapI;

PsatM = 10^(AM - BM/(Y(3) + CM))/760;
PsatI = 10^(AI - BI/(Y(3) + CI))/760;

gM = exp(A/(1 + A*Y(2)/(B*(1-Y(2))))^2);
gI = exp(B/(1 + B*(1-Y(2))/(A*Y(2))))^2);

P1 = (1+A*Y(2)/(B*(1-Y(2))))^3;
P = gM*PsatM*(2*A^2*Y(2)*(1+Y(2)/(1-Y(2)))/(B*(1-Y(2)))/P1-1);

Q1 = (1+B*(1-Y(2))/(A*Y(2)))^3;
Q = gI*PsatI*(2*B^2*(1-Y(2))*(1+(1-Y(2))/Y(2))/(A*Y(2))/Q1-1);

R1 = gM*Y(2)*PsatM*BM/(Y(3)+CM)^2;
R2 = gI*(1-Y(2))*PsatI*BI/(Y(3)+CI)^2;
R = R1 + R2;

dTdxM = (P - Q)/R;

if Y(3) < TBuo
    dNL = 0;
    dxM = 0;
    dT = Qo/(Y(1)*Cp);
else
    dNL = Qo/(Cp*dTdxM*Y(2)*(gM*PsatM/PTot - 1) - Hvap);
    dxM = (gM*PsatM/PTot - 1)*Y(2)*dNL/Y(1);
    dT = dTdxM*dxM;
end

dY = [dNL; dxM; dT];

```

REFERENCES

1. Geankoplis, Christie J., *Transport Processes and Unit Operations*, 2nd Edition, Allyn & Bacon, Boston (1983).
2. Perry, Robert H. and Green, Don W., *Perry's Chemical Engineers' Handbook*, 7th Edition, McGraw Hill, New York (1997).

NOMENCLATURE

Symbol	Definition
A_I, B_I, C_I	Antoine coefficients for isopropanol
A_M, B_M, C_M	Antoine coefficients for methanol
$\underline{C}_{PI}, \underline{C}_{PM}$	Molar heat capacities of isopropanol and methanol, respectively
$\Delta H_{vap\ I}, \Delta H_{vap\ M}$	Molar heat of vaporization for isopropanol and methanol, respectively
K_I, K_M	Kay factors for isopropanol and methanol, respectively
m_i	Weight of the i^{th} instantaneous distillate sample withdrawn from the still
M_o, M_f	Weights of the solution in the still at the start and end of the run, respectively
$M^{TotD}(t)$	Weight of the cumulative distillate collected up till time t , including the n^{th} instantaneous distillate sample
MW_I, MW_M	Molecular weights of isopropanol and methanol, respectively
N_o^L, N_f^L	Total moles of solution in the still at the start and end of the run, respectively
$N^L(t)$	Total moles of solution left in the still at time t
p^{sat}	Saturated vapor pressure
Q_o	A specified (constant) rate of heat supplied to the still in the simulation model
R	Reflux ratio
T_o	Initial temperature of the solution in the still
$w_{M\ o}, w_{M\ f}$	Methanol weight fractions of solution in the still at the start and end of the run, respectively
$w_{M\ i}$	Methanol weight fraction of the i^{th} instantaneous distillate sample
$w_M^{TotD}(t)$	Methanol weight fraction of the cumulative distillate up to time t
x_I	Isopropanol mole fraction of solution in the still
$x_{M\ o}, x_{M\ f}$	Methanol mole fractions of solution in the still at the start and end of the run, respectively
$x_{M\ n}(t)$ or $x_M(t)$	Methanol mole fraction of the solution remaining in the still after the n^{th} instantaneous condensate sample is drawn at time t
$x_M^D(t)$	Methanol mole fraction of the instantaneous distillate sample drawn at time t
$x_{M\ (n)}$	Methanol mole fraction of the liquid stream leaving the n^{th} stage and falling to the $(n+1)^{th}$ stage directly below it
y_I, y_M	Vapor mole fractions of isopropanol and methanol in the still, respectively
$y_{M\ (n+1)}$	Methanol mole fraction of the vapor stream leaving the $(n+1)^{th}$ stage (counting from the top) and rising to the n^{th} stage directly above it
α	One of two constants in the van Laar model for activity coefficients
α_{MI}	Relative volatility for the methanol-isopropanol pair
β	The second constant in the van Laar model for activity coefficients
γ_I, γ_M	Activity coefficients of isopropanol and methanol, respectively

Sample Data Collection Spreadsheet:

Time	Temperature	Mass of Empty Container	Mass of Sample + Container	Mass of Sample	Refractive Index of Sample	Voltage	Watts
1							
2							

Sampling Times: 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 20, 23, 26, 29, 32, 35 minutes